Farhad Foroutan

Robotics & Autonomous Systems Engineer

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Summary

Robotics and Autonomous Systems Engineer with experience across mobile robotics, robotic arms, and embedded IoT. Skilled in autonomous navigation, ROS-based platforms, motion planning, SLAM, computer vision, and sensor fusion. Published researcher in ADAS and advanced control systems. Combines robotics software, control engineering, mechanical design, and field-deployable hardware integration to support autonomous mobility applications.

Experience

Robotics Researcher, Aston University, UK, [04/2025 – Present] **Project Link**

Autonomous Navigation:

- Configured a **Clearpath Jackal** robot as a ROS 2 Humble autonomous platform, integrating SICK TIM 2D LiDAR, IMU, SLAM Toolbox, AMCL, Nav2, RViz, TF, Gazebo, and rosbag2 for mapping, localisation, and goal-based navigation; optimised the stack to **reduce CPU load by 30%.**
- Developed a communication stack for a tethered Jackal-drone system, using **CAN bus** and **serial communication** for real-time pose and localisation transmission and coordination with embedded winch hardware.

Multi-Camera Data Collection (SocNav):

- Developed a **ROS 2 Humble 360° human perception pipeline** for an Insta360 camera, integrating YOLO-based person detection, multi-frame tracking, equirectangular-to-perspective processing, and pose estimation to provide real-time human localisation, orientation, and upper-body key points for downstream robotic interaction.
- Built a roof-mounted portable recording system integrating 8 ELP fisheye cameras, 4 Jetsons (AGX/Xavier), a network switch, and external power supply for synchronised 360° video capture, achieving **sub-2 millisecond inter-camera timestamp offset**.
- Developed a **11-step autonomous data-capture pipeline**, linking robot RViz goal input, Nav2 execution, SSH-based Jetson triggering, and synchronised rosbag2 recording together.

Quadruped & Manipulation Robotics:

- Programmed a smart courier agent on a **Unitree GO2-W quadruped** for safe hospital navigation and independent elevator interaction; implemented real-time image processing (Python, OpenCV, Scikit-image) for button detection, achieving **90%+ task accuracy**.

Research Assistant, Wearable IMU Glove for Clinical Hand-Function Assessment, Aston University, [09/2026 – Present]

- Developed a Python pipeline estimating finger range of motion from a 6-IMU wearable glove, processing a dataset of 15,000+ hand-function recordings from human participants.
- Designed a **PCA-based** technique that derives each finger's functional rotation axis from quaternion data instead of the IMU's main axis, enabling fewer sensors than comparable gloves while **capturing on average 98% of finger motion**.

Embedded IoT Engineer, UKRI-Funded Project, Aston University, UK, [11/2025 – 07/2026] **Project Link**

- Developed a **Python-based automation platform** for cold-chain telemetry, integrating 9 heterogeneous sensors spanning refrigeration, weather, GPS, and mission-route data into a unified pipeline with structured JSON modelling and 5-second sampling.
- Engineered an automated cloud-sync workflow across **1-Wire, I2C, UART, GPIO**, **and 4G LTE**, automatically uploading structured telemetry to cloud storage every 30 seconds via SIM7600G-H and rclone for continuous visibility into distributed field-deployed equipment.
- Deployed the system as a headless Linux service using systemd auto-start, crash recovery, and self-healing logic, **enabling zero-touch, always-on operation.**
- Built modular, config-driven Python pipelines for parsing and validating multi-source data streams, enabling new sensor types and data formats to be added without modifying core logic; implemented validation and error handling to maintain pipeline stability against malformed or missing readings.
- Contributed to the design of a **SQL-based telemetry database** for a companion web platform, developing a schema to archive and query structured sensor and mission-route data for reporting and analysis.

Mechanical Engineering Intern, Iran (Seasonal, 2020–2024)

- ***Faramashin Corporation, Komatsu Iran****, 2024:* Maintained mechanical and electrical systems on heavy mining machinery (bulldozers, dump trucks, shovels); contributed to an in-house CNC team producing 15 components/week.
- ***IKCO R&D Division****, 2022:* Performed force analysis and FEM simulation of 4 progressive automotive moulds for Dena components using ANSYS and Abaqus to support R&D validation.
- ***Tabkh Shamim Industries****, 2020-2021:* Hands-on operation of SMAW and TIG welding, CNC machining, and sheet metal forming across multiple product lines including vegetable washers and cutlery polishers; produced technical drawings for manufacturing.

Research Assistant, Dynamics and Control Lab, Shiraz University, [04/2022 – 08/2024] **Google Scholar Link**

- Published **2 peer-reviewed papers** in self-driving vehicles and advanced control systems; research covered ADAS, motion planning, sensor fusion, vehicle dynamics, and sustainable energy management.
- Developed a novel **Adaptive Cruise Control (ACC)** algorithm in MATLAB/Simulink, achieving **10% fuel efficiency improvement** through lateral motion-aware passenger comfort optimisation.

Start-Up Founder, Everlum, [03/2024 – Present], (**www.everlum.ir** | Instagram: @everlum.ir)

- Co-founded a premium metal poster start-up with two product lines; **managed a 9-member team** across production, packaging, and digital marketing, securing partnerships with major Iranian companies and achieving consistent sales growth.
- Achieved **50K+ annual sales** and grew the brand's Instagram following to **192K within 18 months** through targeted content strategy and community engagement.

Core Skills

- **Programming & Modelling:** Python, C++, Bash, SQL, MATLAB/Simulink, Git/GitHub, OOP
- **Robotics & Perception:** ROS (1&2), SLAM Toolbox, AMCL, Nav2, RViz, TF, rosbag, OpenCV, Scikit-image, YOLO, CoMotion, Gazebo, Isaac Sim
- **Embedded & Hardware:** NVIDIA Jetson, Raspberry Pi, Arduino, ARM, CAN bus, UART, I2C, 1-Wire, GPIO, 4G LTE
- **Linux, Networking & DevOps:** Linux, Docker, SSH, systemd, Netplan, NetworkManager, rclone, GCP
- **Robotic Platforms**: Clearpath Jackal, Unitree GO2-W, PAL TIAGo Pro, Kinova arms, Dobot arms, Festo Robotino
- **Simulation, CAD & Engineering:** Fusion360, SolidWorks, ANSYS (Fluent, Mechanical, APDL), Abaqus
- **Standards:** ETSI EN 303 645, ISO 3691-4, UL 4600, SAE J3016
- **Languages:** English, Persian

Education

M.Sc. – Robotics and Autonomous Systems, Aston University, UK, [04/2025 – 04/2026]

- **Merit Scholarship; Best Dissertation Award**.
- **Thesis:** *Design and development of a mobile multi-modal data acquisition platform for social robot navigation benchmarking*.

B.Sc. – Mechanical Engineering, Shiraz University, Iran, [09/2019 – 06/2023]

- Graduated in the **top 10%** of the class; member of the Exceptional Talent organisation.
- **Thesis:** *A new adaptive cruise control strategy considering road conditions.*

Teaching & Selected Projects

Teaching:

- Supported teaching for *Advanced Dynamics and Control*, *Signals and Systems*, and *Production Design 1 &2*; delivered laboratory instruction for Fluid Mechanics; mentored 2 MSc students.

Selected Academic Projects:

- Designed a MATLAB/Simulink closed-loop thermal control system with a 6×8 spatial sensing grid, distributed zonal PID control, and adaptive gain scheduling, **reducing sensing interconnections by 70%**.
- Built an **Arduino-based autonomous sailboat** integrating GPS navigation, compass heading, wind sensing, and closed-loop rudder/sail actuation; validated through real-water deployment with onboard SD-card telemetry logging.